

# Memory and CPUs Assigning to VMs in order to obtain maximum performance according to numa topology

 **Horacio Marchioni** August 19, 2019 in [Guest Runtime](#)

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 **Horacio Marchioni** Legacy Group  
August 19, 2019



Hello, I am working with Xenserver 7.1

I need to control how different VMs are loaded in different NUMA nodes at start up. I have different servers where I want to implement this, IBM, HP and Dell.

I had found some information about the CLI commands in [https://wiki.xenproject.org/wiki/Cpupools\\_Howto](https://wiki.xenproject.org/wiki/Cpupools_Howto) and also several links inside that link showing how to improve performance, but I couldn't found exactly how, where and in which files in xen 7.1 I need to include the commands to achieve this when starting up the server. I have also tried with those CLI commands with the server running, but I could not get my VMs to be loaded in an specific NUMA node with all its memory from that NUMA node to attain best performance possible..

Thank in advance, I would appreciate any help on this issue

Horace

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**Tobias Kreidl** Members  
August 19, 2019



Horace,

For starters, I'd suggest checking out <https://www.mycugc.org/blogs/tobias-kreidl/2019/04/30/a-tale-of-two-servers-part-3>

and some of the references contained therein, as well as various older Xen documents that discuss various means of loading VMs to optimize NUMA and possibly also looking into VCPU pinning. For a Xen environment, which will be similar enough to XenServer to be relevant, see for example: [https://wiki.xen.org/wiki/Xen\\_on\\_NUMA\\_Machines](https://wiki.xen.org/wiki/Xen_on_NUMA_Machines)

Wither or order of starting VMs has to be respected or CPU pinning to ensure reasonable NUMA optimization. It is also important to consider the number of VCPUs to assign as well as how much memory to allocate as part of the optimization process, and those choices will of course depend on the server architecture involved.

--Tobias

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